

F1: Design Viability Judgment

In order to begin construction of our portable sink, we must first identify the possible hazards and threats of our construction process.

Items that could potentially cause issues and why they we can mitigate this:

- Over budget project: This is one of the most important issues to not have happen. For our project we only have a total budget of about \$200 for this project of outside materials that we do not already have. In order to prevent this issue we are taking steps to calculate the cost of each item we expect to buy in order to build the project and leaving a little bit of space so that if there is an unexpected purchase needed we have room to add in the extra cost. Further, while our calculated bill of materials exceeds \$200 (+\$16.57), we have sourced the majority of our materials from in-school sources, offering them free of charge(*Note: the final bill of materials includes the estimated price the materials we sourced for free would cost if we had to purchase these items*). Furthermore, we are 100% confident that the true cost of the project was below \$200 as a majority of our materials were in-house sourced, likely being worth more than the exceeded \$16.57.
- Water being too slow: This problem can make or break the sink. If we run into this problem our sink will be at a severe disadvantage to the teacher and students using the regular sink as if you cannot perform the actions of washing your hands and brushing your teeth than a sink in the context of a life skills class will not be useful. In order to prevent this issue we have a valve that will control the speed of the water and allow more to flow if needed. We also have the ability to widen the hole of the water storage if needed as the storage container we are using has 2 holes with one being bigger than the other to allow a faster flow. Further, we have also experimented with the inclusion of multiple faucets, with different size holes, allowing for different amounts of water to flow. It is safe to say that our prototype will be able to be adjusted by the stakeholder to ensure the waterflow is not too slow. On the flip side, the sink's ability to use water from two tanks at a time may allow for a water flow too strong for proper use of the sink. This is also mitigated by the different faucets with different size holes, allowing the stakeholder to control the flow outside of just the sink handles adjustability.
- Contamination and mold growth: Considering the fact that we are using an appliance that provides water for brushing teeth and washing hands, it is very important that we use a sink that can be easily cleaned, and has no space favorable to bacteria growth. Conveniently, our sink's entire surface will be constructed out of plastic. This will facilitate easy cleaning via wipes, and no nooks or crannies which could allow bacteria growth. Further when examining where the water makes contact with our sink, we've determined there only being one potential area the bacterial growth may need to be cleaned to avoid further deterioration of the sink. According to Johnathan T Blanton, "stainless steel is the best material at resisting bacterial growth." Further we acknowledge

that water makes contact with the stainless steel sink bowl, as well as the internal compartment of the large plastic storage container. Moreover, the concern that the bacterial growth in the sink is minimal, and should not draw bacterial growth due to the frequency the sink will be used/flushed, as well as the materials ability to withsWith proper instructions as to how to disinfect this internal compartment once a week should ensure that our prototype mitigates the cleanliness concerns of the stakeholders, experts, and possible investors.

-
- *Household Surfaces and Bacteria*, csef.usc.edu/History/2007/Projects/J1404.pdf. Accessed 17 May 2024.
-
-
- Spillage: When working with water it is an usual occurrence for water to spill out and make a mess. This is not ideal so it would be critical if we could prevent the spillage of the water preventing messes in the classroom. In order to achieve this goal we are utilizing the lid of the container to act as a lid on the sink that has edges to prevent spillage. Another preventing measure is that we have more than enough room in the bottom of our sink holding the gray water allowing enough space to hold the wastewater and its contents without flow coming back up through the sink. Further our use of water-proof grade water containers, allows us to ensure that this is not an issue. Moreover, when looking at the sink faucet and handles, we have mitigated possible leaking through the use of superglue to attach each piece. This will surprisingly mitigate water leakage, as large amounts of the glue act as a sturdy solid barrier between the liquid and external surfaces.
- Stability: One issue that could happen is that the sink will tip due to the weight being mainly put on one edge bottom container. This issue is rational and could happen if we are not careful with the way we place our clean water storage. The way we mitigate this is by proper placement of our clean water storage, we are expecting to put it on the top of the storage bin's lid but having it take up at least half of the lid on one side in order to make sure the weight is distributed on each side. Another thing we can do to mitigate unstableness is to add a counterweight or just weight into the container in order to make the bottom more heavy than the top; however this could cause issues with overall weight but we would just have to monitor the overall weight as we go to make sure that the sink is still able to be picked up by an average person and the teacher more importantly. Further, knowing we cannot work around the fact that the sink requires a waste storage system as well as a water storage system, we understand the height and weight of the sink can only be reduced and mitigated so much. Moreover, in the case it causes problems, we have decided to recommend to the teachers to ensure the sink is in an area that may allow it to rest next to the wall, to reduce the chance a student runs by and knocks it over.

- Lack of functionality or learning capabilities: One of the aspects of the projects that makes this project engineering is the fact that we are making a portable sink and the fact that it is being used in a classroom so a learning element should be included as well. One issue we run into is the lack of having learning capabilities or the lack of functionality. These two go hand in hand in making our sink so it is important for us to find a good balance between the two. The most important of these 2 is the functionality of the sink. However, adding in some learning capabilities will be excellent and will make our design go much further in pleasing the customer. Our main concern in our design is functionality, but we are deciding to add a learning capability if possible in the design. Our ideas for adding in a learning feature is the addition of 3D printed sink handles that model ones that are commonly used in the real world. This will allow students lacking motor skills the ability to have experience and figure out how to operate different types of handles that will appear as they traverse life. We also have decided to color code the temperatures of the waters, to add a learning lesson in terms of which side gives what temperature of water, and what the purpose is to have two sink handles.